2.6 The Arithmetic–Geometric Mean

WORKSHEET

The question

STEP II 2024, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Why is the arithmetic mean of two different positive numbers always bigger than their geometric mean? Can you prove it in one line?

STAGE 1 The two sequences squeeze together

Ask yourself Why is the arithmetic mean of x_n and y_n less than x_n, and the geometric mean bigger than y_n?

STAGE 2 The substitution

Ask yourself With $t = \frac{1}{2}\left(x - \frac{pq}{x}\right)$, compute $t^2 + pq$. Then compute $t^2 + \frac{1}{4}(p+q)^2$. Do either of them factorise nicely?

Pause and try Put these together. As x runs from 0 to ∞, what does t do? What does the t-integral become?

STAGE 3 Evaluating the integral

Ask yourself Apply this identity repeatedly, starting from (x_0, y_0). What is $\mathrm{I}(x_0, y_0)$ equal to after n steps? What happens as $n \to \infty$?

Try this next

1. **Quadratic convergence.** Prove the identity $x_{n+1} - y_{n+1} = \dfrac{(x_n - y_n)^2}{2(\sqrt{x_n} + \sqrt{y_n})^2}$, and deduce that $x_{n+1} - y_{n+1} \le \dfrac{(x_n - y_n)^2}{8y_0}$.

2. **By hand.** Compute $M(2,1)$ to six decimal places using a calculator. How many steps did you need?

3. **Scaling.** Show that $M(\lambda x, \lambda y) = \lambda M(x,y)$ for $\lambda > 0$, and that $\mathrm{I}(\lambda p, \lambda q) = \lambda^{-1}\mathrm{I}(p,q)$. Check that these two facts are consistent with $\mathrm{I}(x_0, y_0) = \pi/(2M)$.

4. **The pendulum.** Use the formula above to find, to three significant figures, how much longer the period of a pendulum released from 60° is than the small-angle prediction.

Further reading

- **David Cox, "The arithmetic–geometric mean of Gauss"**, *L'Enseignement Mathématique* 30 (1984). The classic account of the history and mathematics; the first sections need only calculus.
- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). The definitive book on the AGM and fast computation of π.
- **Jonathan Borwein & Keith Devlin, *The Computer as Crucible*** (A. K. Peters). Experimental mathematics, including discoveries like Gauss's, made by spotting numerical coincidences.

The full guided solution, and what the question leads to at university, is in Section 2.6:
https://beyondstep.keithmcnulty.org/ch02/agm.html